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DISPLAY PANEL AND DISPLAY APPARATUS

Abstract

Embodiments of the present application disclose a display panel and a display apparatus. A light transmittance of the first active area of the display panel is greater than a light transmittance of the second active area of the display panel. The display panel includes a plurality of first pixel units located in the first active area and a plurality of first pixel circuits located in the second active area. The first pixel unit includes a plurality of sub-pixels. The first pixel circuit is connected to the first electrode of the corresponding sub-pixel in the first pixel unit through a wire. The first pixel circuits are arranged in an array. The first pixel circuits corresponding to the plurality of sub-pixels in the same first pixel unit are arranged in at least two rows and in a number of columns less than a total number of sub-pixels in the first pixel unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of International Application No. PCT/CN2023/101857 filed on Jun. 21, 2023, which claims priority to Chinese Patent Application No. 202211537308.0, filed with the China National Intellectual Property Administration on Dec. 1, 2022. All of the aforementioned patent applications are hereby incorporated by reference in their entireties.

FIELD

[0002] Embodiments of the present application relate to the field of display technologies, and for example, to a display panel and a display apparatus.

BACKGROUND

[0003] As display technologies continuously develop, people have increasingly high requirements for a screen-to-body ratio of display apparatuses, and full-screen display technologies have become the mainstream research trend in the future. Currently, an under display camera (UDC) technology has been applied in the display apparatuses. In the technology, a display screen is divided into a primary-screen area and a secondary-screen area. Both the primary-screen area and the secondary-screen area are used for display, and the secondary-screen area is a light-transmissive area in which an under display camera is provided. The secondary-screen area needs to meet both a requirement for a relatively high light transmittance and a requirement for display effects. However, the display apparatuses usually have a problem of poor display effects in secondary-screen areas.

SUMMARY

[0004] Embodiments of the present application provide a display panel and a display apparatus, to improve the display effect of a first active area of the display panel and mitigate the problem of display non-uniformity in the first active area.

[0005] According to a first aspect, an embodiment of the present application provides a display panel. The display panel has a first active area and a second active area, where a light transmittance of the first active area is greater than a light transmittance of the second active area. The display panel includes: [0006] a plurality of first pixel units located in the first active area, the first pixel unit including a plurality of sub-pixels, where the sub-pixel includes a first electrode and a second electrode opposite the first electrode; and [0007] a plurality of first pixel circuits located in the second active area, the first pixel circuit being connected to the first electrode of the corresponding sub-pixel in the first pixel unit through a wire, [0008] where the plurality of first pixel circuits are arranged in an array, and the first pixel circuits corresponding to the sub-pixels in a same first pixel unit are arranged in at least two rows and in a number of columns less than a total number of sub-pixels in the first pixel unit.

[0009] According to a second aspect, an embodiment of the present application provides a display apparatus, including a display panel according to the first aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** is a schematic diagram of a structure of a display panel in the related art;

[0011] FIG. **2** is a schematic diagram of a structure of a display panel according to an embodiment of the present application;

[0012] FIG. **3** is a partial enlarged view of an area B**1** in FIG. **2**;

[0013] FIG. **4** is another partial enlarged view of the area B**1** in FIG. **2**;

[0014] FIG. **5** is a partial enlarged view of an area B**2** in FIG. **2**;

[0015] FIG. **6** is another partial enlarged view of the area B**2** in FIG. **2**;

[0016] FIG. **7** is a schematic diagram of a structure of another display panel according to an embodiment of the present application;

[0017] FIG. **8** is a partial enlarged view of an area D**1** in FIG. **7**; and

[0018] FIG. **9** is a schematic diagram of a structure of another display panel according to an embodiment of the present application.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0019] Display apparatuses have a problem of poor display effects in secondary-screen areas. Through research, it has been found that the reason for the above problem is as follows.

[0020] FIG. **1** is a schematic diagram of a structure of a display panel in the related art. Referring to FIG. **1**, for example, the display panel includes an active area **100**. The active area **100** includes a primary-screen area **01**, a secondary-screen area **02**, and a transition area **03** located between the primary-screen area **01** and the secondary-screen area **02**. The primary-screen area **01**, the secondary-screen area **02**, and the transition area **03** each include a plurality of sub-pixels to achieve a display function, and the secondary-screen area **02** is a light-transmissive area in which an under display camera is provided. The secondary-screen area **02** needs to meet a requirement for relatively high light transmittance to achieve a relatively low diffraction effect, and also needs to meet a requirement for display effects. Therefore, pixel circuits that drive the sub-pixels of the secondary-screen area **02** need to be arranged in the transition area **03** around the secondary-screen area **02**, to improve the light transmittance of the secondary-screen area **02** and reduce the diffraction effect of the secondary-screen area **02**. However, the transition area **03** is usually a relatively large area, which makes connection lines between the pixel circuits in the transition area **03** and the sub-pixels in the secondary-screen area **02** excessively long, resulting in relatively large parasitic capacitances in the connection lines and slow charging rates at anodes of light-emitting devices of the sub-pixels in the secondary-screen area **02**, and thus affecting the display effect in the secondary-screen area **02**. Moreover, the connection lines between the sub-pixels at different positions in the secondary-screen area **02** and the corresponding pixel circuits in the transition area **03** have different lengths, which makes different parasitic capacitances in different connection lines and also different charging rates at anodes of light-emitting devices of different sub-pixels, resulting in the problem of display non-uniformity easily occurring in the secondary-screen area **02** at a low gray level.

[0021] Embodiments of the present application provide a display panel. FIG. **2** is a schematic diagram of a structure of a display panel according to an embodiment of the present application. FIG. **3** is a partial enlarged view of an area B**1** in FIG. **2**. Referring to FIG. **2** and FIG. **3**, the display panel has an active area AA and a non-active area NAA. The active area AA includes a first active area AA**1** and a second active area AA**2**, where a light transmittance of the first active area AA**1** is greater than a light transmittance of the second active area AA**2**. The display panel includes a plurality of first pixel units **10** and a plurality of first pixel circuits **30**. The first pixel units **10** are located in the first active area AA**1**. The first pixel unit **10** includes a plurality of sub-pixels **110**. The sub-pixel **110** includes a first electrode and a second electrode opposite the first electrode. The first pixel circuits **30** are located in the second active area AA**2**. The first pixel circuit **30** is connected to the first electrode of the corresponding sub-pixel **110** in the first pixel unit **10** through a wire **40**.

[0022] The plurality of first pixel circuits **30** are arranged in an array. The first pixel circuits **30** corresponding to the plurality of sub-pixels **110** in the same first pixel unit **10** are arranged in at least two rows and in a number of columns less than a total number of sub-pixels **110** in the first pixel unit **10**.

[0023] For example, the display panel according to the embodiment of the present application may be an organic light-emitting diode (OLED) display panel, an active-matrix organic light-emitting diode (AMOLED) display panel, a micro light-emitting diode (micro-LED) display panel, etc. The first active area AA**1** includes a plurality of first pixel units **10**. The second active area AA**2** includes a plurality of second pixel units **20**. The first pixel unit **10** and the second pixel unit **20** each include a plurality of sub-pixels **110**. The sub-pixel **110** may include a light-emitting device, such as an OLED or a micro LED. In a possible case, the first electrode of the sub-pixel may be an anode of the light-emitting device, and the second electrode of the sub-pixel may be a cathode of the light-emitting device. In other possible cases, the first electrode of the sub-pixel may be the cathode of the light-emitting device, and the second electrode of the sub-pixel may be the anode of the light-emitting device.

[0024] The first pixel circuits **30** are configured to drive the corresponding sub-pixels **110** in the first pixel unit **10** to emit light. The display panel further includes a plurality of second pixel circuits (not shown in FIG. **2** and FIG. **3**). The second pixel circuits are located in the second active area AA**2**, and are configured to drive the corresponding sub-pixels **110** in the second pixel unit **20** to emit light. The sub-pixels **110** in the first active area AA**1** are driven by the first pixel circuits **30** to emit light, and the corresponding sub-pixels **110** in the second active area AA**2** are driven by the second pixel circuits to emit light, such that the display panel can implement a display function. The second active area AA**2** may be a normal active area, which is also referred to as a primary-screen area. The first active area AA**1** may be a transparent active area, which is also referred to as a secondary-screen area. The display panel has an active side and a non-active side opposite the active side. On the non-active side corresponding to the first active area AA**1**, a photosensitive element may be arranged. For example, the photosensitive element may be an under display camera. In a display mode, both the first active area AA**1** and the second active area AA**2** display light normally. In a photographing mode, the first active area AA**1** has a relatively high light transmittance, which enables light to pass through the first active area AA**1** into the under display camera, such that the under display camera can sense light and then perform photographing.

[0025] The first pixel circuit **30** includes a driving current output terminal. The driving current output terminal of the first pixel circuit **30** may be connected to the anode of the light-emitting device of the corresponding sub-pixel **110** through the wire **40**, to supply a driving current to the anode of the light-emitting device to drive the light-emitting device to emit light. A plurality of sub-pixels **110** in the same first pixel unit **10** may correspond to a plurality of first pixel circuits **30**. Configuring the first pixel circuits **30** corresponding to the plurality of sub-pixels **110** in the same first pixel unit **10** to be arranged in at least two rows and in a number of columns less than the total number of sub-pixels **110** in the first pixel unit **10** enables at least a portion of the first pixel circuits **30** to be arranged in the same column to reduce the number of columns in which the plurality of first pixel circuits **30** are arranged, so as to reduce widths of the plurality of first pixel circuits **30** in a row direction in which the plurality of first pixel circuits are arranged, reduce distances between the first pixel circuits **30** and the sub-pixels **110** and then reduce lengths of the wires **40** connected between the first pixel circuits **30** and the sub-pixels **110**. Reducing the lengths of the wires **40** facilitates reducing the parasitic capacitances in the wires **40**, to increase charging rates of the first pixel circuits **30** for the anodes of the light-emitting devices of the sub-pixels **110**, thereby ensuring the display effect in the first active area AA**1**. Since the lengths of all of the wires **40** between the plurality of sub-pixels **110** in the first active area AA**1** and the corresponding first pixel circuits **30** are reduced, the embodiment facilitates reducing the difference in lengths between the wires **40** connected to the plurality of sub-pixels **110** to reduce the difference in charging rates between the anodes of the light-emitting devices of the plurality of sub-pixels **110**. Therefore, the problem of display non-uniformity in the first active area AA**1** is mitigated.

[0026] For example, when the first pixel unit **10** includes eight sub-pixels **110** and the eight sub-pixels **110** correspond to a total of eight first pixel circuits **30**, the corresponding eight first pixel circuits **30** may be configured to be arranged in at least two rows and in less than eight columns. For example, the eight first pixel circuits **30** may be configured to be arranged in two rows and four columns. Compared with the solution in which the eight first pixel circuits **30** are arranged in one row and eight columns, in the embodiment, the number of columns in which the plurality of first pixel circuits **30** are arranged is reduced, so that the lengths of the wires **40** between at least a portion of the first pixel circuits **30** and the corresponding sub-pixels **110** can be reduced, to reduce the parasitic capacitances in the wires **40** and improve charging rates of the first pixel circuits **30** for the anodes of the light-emitting devices of the sub-pixels **110**, thereby ensuring the display effect in the first active area AA**1**. In addition, this further facilitates reducing the difference in lengths between the wires **40** connected to the plurality of sub-pixels **110**. Therefore, the problem of display non-uniformity in the first active area AA**1** is mitigated.

[0027] To sum up, in the embodiment of the present application, the first pixel circuits corresponding to the plurality of sub-pixels in the same first pixel unit are configured to be arranged in at least two rows and in a number of columns less than the total number of sub-pixels in the first pixel unit. This facilitates reducing the number of columns in which the plurality of first pixel circuits are arranged, to reduce distances between the first pixel circuits and the sub-pixels, and then reduce lengths of wires connected between the first pixel circuits and the sub-pixels, thereby improving charging rates of the first pixel circuits for the sub-pixels. Therefore, the display effect in the first active area is ensured. In addition, since the lengths of all of the wires connected to the plurality of sub-pixels in the first active area are reduced, the embodiments of the present application further facilitate reducing the difference in lengths between the wires connected to the plurality of sub-pixels, to reduce the difference in charging rates between the plurality of sub-pixels. Therefore, the problem of display non-uniformity in the first active area is mitigated.

[0028] It should be noted that FIG. **3** only shows that the first pixel circuits **30** corresponding to the plurality of sub-pixels **110** in the same first pixel unit **10** are arranged in two rows. In other embodiments, the first pixel circuits **30** corresponding to the plurality of sub-pixels **110** in the same first pixel unit **10** may alternatively be arranged in more than two rows, to further reduce the number of columns in which the plurality of first pixel circuits **30** are arranged, thereby reducing lengths of the wires **40** between the first pixel circuits **30** and the corresponding sub-pixels **110**.

[0029] Referring to FIG. **2** and FIG. **3**, on the basis of the above embodiments, the first pixel unit **10** may be configured to include m sub-pixels **110** of the same color, and the m sub-pixels **110** of the same color in the same first pixel unit **10** are driven by n first pixel circuits **30**, where n and m are both positive integers greater than or equal to 1, and $n \leq m$. For example, for $n=m=1$, one first pixel circuit **30** can drive one sub-pixel **110** in the first pixel unit **10**, enabling one driving one. For $n=1$ and $m>1$, one first pixel circuit **30** can drive a plurality of sub-pixels **110** of the same color in the first pixel unit **10**, enabling one driving many. For $n>1$ and $m>1$, a plurality of first pixel circuits **30** can drive a plurality of sub-pixels **110** of the same color in the first pixel unit **10**, enabling many driving many.

[0030] In one embodiment, for $n=m=1$, each first pixel circuit **30** is connected to one corresponding sub-pixel **110** through a wire **40**. For $n=1$ and $m>1$, the first electrodes of the m sub-pixels **110** of the same color in the same first pixel unit **10** are electrically connected to each other, and are electrically connected to the corresponding first pixel circuits **30** through the same wire **40**. For $n>1$ and $m>1$, n first pixel circuits **30** corresponding to the m sub-pixels **110** of the same color in the same first pixel unit **10** are electrically connected to each other, and the first electrodes of the m sub-pixels **110** of the same color are electrically connected to each other, and are electrically connected to any one first pixel circuit **30** among the n corresponding first pixel circuits **30** through the same wire **40**.

[0031] For example, the sub-pixel **110** includes a light-emitting device. The wire **40** is connected between a driving current output terminal of the first pixel circuit **30** and the first electrode (for example) of the light-emitting device in the corresponding sub-pixel **110**. For n=m=1, the driving current output terminal of each first pixel circuit **30** is connected to the first electrode of the light-emitting device in one corresponding sub-pixel **110** through the wire **40**. For n=1 and m>1, the first electrodes of the light-emitting devices in the m sub-pixels **110** of the same color in the same first pixel unit **10** are electrically connected to each other, and are connected to the driving current output terminal of one corresponding first pixel circuit **30** through the same wire **40**. For n>1 and m>1, the driving current output terminals of the n first pixel circuits **30** corresponding to the m sub-pixels **110** of the same color in the same first pixel unit **10** are electrically connected to each other, and the first electrodes of the m sub-pixels **110** of the same color are electrically connected to each other, and are electrically connected to the driving current output terminal of any first pixel circuit **30** among the n corresponding first pixel circuits **30** through the same wire **40**. In one embodiment, for n>1 and m>1, the m sub-pixels **110** of the same color in the same first pixel unit **10** are connected to the closest first pixel circuit **30** among the n corresponding first pixel circuits **30** through the wire **40**. Such a configuration has an advantage of reducing the length of the wire **40** to reduce the parasitic capacitance in the wire **40**, so as to increase the charging rates of the first pixel circuits **30** for the anodes of the light-emitting devices of the sub-pixels **110**, thereby improving the display effect in the first active area AA**1**.

[0032] In one embodiment, the plurality of sub-pixels **110** in the first pixel unit **10** have different colors, and there is at least one sub-pixel **110** of each color. In an embodiment, the first pixel circuits **30** corresponding to the sub-pixels **110** of the same color in the same first pixel unit **10** may be configured to be arranged in the same column.

[0033] For example, the first pixel unit **10** includes a first sub-pixel **111**, a second sub-pixel **112**, and a third sub-pixel **113**. The first sub-pixel **111**, the second sub-pixel **112**, and the third sub-pixel **113** have different colors. The first pixel circuit **30** includes a first sub-pixel circuit **310**, a second sub-pixel circuit **320**, and a third sub-pixel circuit **330**. The first sub-pixel circuit **310** is configured to drive the first sub-pixel **111**, the second sub-pixel circuit **320** is configured to drive the second sub-pixel **112**, and the third sub-pixel circuit **330** is configured to drive the third sub-pixel **113**. The first sub-pixel circuits **310** corresponding to the sub-pixels **110** of the same color in the same first pixel unit **10** may be arranged in the same column, the second sub-pixel circuits **320** corresponding to the sub-pixels **110** of the same color in the same first pixel unit **10** may be arranged in the same column, and the third sub-pixel circuits **330** corresponding to the sub-pixels **110** of the same color in the same first pixel unit **10** may also be arranged in the same column. Such a configuration has an advantage of reducing the number of columns in which the first pixel circuits **30** are arranged, to reduce the widths of the first pixel circuits **30** in the row direction in which the first pixel circuits are arranged, reduce the distances between the first pixel circuits **30** and the sub-pixels **110**, and then reduce the lengths of the wires **40** connected between the first pixel circuits **30** and the sub-pixels **110**. Moreover, arranging the first pixel circuits **30** corresponding to the sub-pixels **110** of the same color in the same column further facilitates simplifying the layout of connection lines between the first pixel circuits **30** corresponding to the sub-pixels **110** of the same color.

[0034] In other embodiments, the first pixel circuits **30** corresponding to the sub-pixels **110** of the same color in the same first pixel unit **10** may alternatively be configured to be arranged in the same row. For example, the first sub-pixel circuits **310** corresponding to the sub-pixels **110** of the same color in the same first pixel unit **10** may be arranged in the same row, the second sub-pixel circuits **320** corresponding to the sub-pixels **110** of the same color in the same first pixel unit **10** may be arranged in the same row, the third sub-pixel circuits **330** corresponding to the sub-pixels **110** of the same color in the same first pixel unit **10** may be arranged in the same row, and the first pixel circuits **30** corresponding to sub-pixels **110** of different colors may be arranged in the same or different rows. Such a configuration has an advantage of simplifying the layout of the connection lines between the first pixel circuits **30** corresponding to the sub-pixels **110** of the same color.

[0035] Referring to FIG. **3**, in one embodiment, the same first pixel unit **10** includes two first sub-pixels **111** and two third sub-pixels **113**, as well as four second sub-pixels **112**. The first sub-pixel **111**, the second sub-pixel **112**, and the third sub-pixel **113** are arranged in an array in the first pixel unit **10**. The first sub-pixel **111**, the second sub-pixel **112**, the third sub-pixel **113**, and the second sub-pixel **112** are sequentially arranged in a first row of the first pixel unit **10**, and the third sub-pixel **113**, the second sub-pixel **112**, the first sub-pixel **111**, and the second sub-pixel **112** are sequentially arranged in a second row of the first pixel unit **10**. In one embodiment, the first sub-pixel **111** is a red sub-pixel **110**, the second sub-pixel **112** is a green sub-pixel **110**, and the third sub-pixel **113** is a blue sub-pixel **110**.

[0036] In one embodiment, the two first sub-pixels **111** in the same first pixel unit **10** may be configured to be driven by two first sub-pixel circuits **310**, the two third sub-pixels **113** in the same first pixel unit **10** may be configured to be driven by two third sub-pixel circuits **330**, and the four second sub-pixels **112** in the same first pixel unit **10** may be configured to be driven by four second sub-pixel circuits **320**. In the first pixel circuits **30** corresponding to the same first pixel unit **10**: two first sub-pixel circuits **310** are arranged in the same column, two third sub-pixel circuits **330** are arranged in the same column, four second sub-pixel circuits **320** are arranged in two columns, and two first sub-pixel circuits **310**, four second sub-pixel circuits **320**, and two third sub-pixel circuits **330** are arranged in at least two rows.

[0037] For example, the first sub-pixel circuit **310**, the second sub-pixel circuit **320**, and the third sub-pixel circuit **330** corresponding to the plurality of sub-pixels in the same first pixel unit **10** may be arranged in two rows, where each row includes the first sub-pixel circuit **310**, the second sub-pixel circuit **320**, and the third sub-pixel circuit **330**, and the first pixel circuits **30** for driving the sub-pixels of the same color are located in the same column. Such a configuration has an advantage of reducing the number of columns in which the first pixel circuits **30** are arranged, to reduce the widths of the first pixel circuits **30** in the row direction in which the first pixel circuits are arranged, reduce the distances between the first pixel circuits **30** and the sub-pixels **110**, and then reduce the lengths of the wires **40** connected between the first pixel circuits **30** and the sub-pixels **110**. Moreover, arranging the first pixel circuits **30** corresponding to the sub-pixels **110** of the same color in the same column further facilitates simplifying the layout of connection lines between the first pixel circuits **30** corresponding to the sub-pixels **110** of the same color.

[0038] Still referring to FIG. **3**, one column of first sub-pixel circuits **310**, two columns of second sub-pixel circuits **320**, and one column of third sub-pixel circuits **330** are sequentially arranged on one side of the corresponding first pixel unit **10**. This facilitates reducing the distance between the first sub-pixel circuit **310** and the first sub-pixel **111** to reduce the length of the wire **40** between the first sub-pixel circuit **310** and the first sub-pixel **111**, and facilitates reducing the distance between the second sub-pixel circuit **320** and the second sub-pixel **112** to reduce the length of the wire **40** between the second sub-pixel circuit **320** and the second sub-pixel **112**.

[0039] In other embodiments, one column of first sub-pixel circuits **310**, one column of third sub-pixel circuits **330**, and two columns of second sub-pixel circuits **320** may alternatively be configured to be sequentially arranged on one side of the corresponding first pixel unit **10**. This facilitates reducing the distance between the first sub-pixel circuit **310** and the first sub-pixel **111** to reduce the length of the wire **40** between the first sub-pixel circuit **310** and the first sub-pixel **111**, and facilitates reducing the distance between the third sub-pixel circuit **330** and the third sub-pixel **113** to reduce the length of the wire **40** between the third sub-pixel circuit **330** and the third sub-pixel **113**.

[0040] FIG. **4** is another partial enlarged view of the area B**1** in FIG. **2**. Referring to FIG. **4**, in another embodiment, the first pixel circuits **30** corresponding to the sub-pixels **110** of the same color in the same first pixel unit **10** may be configured to be arranged in different rows and/or different columns. For example, the first pixel circuits **30** corresponding to sub-pixels **110** of different colors in the same first pixel unit **10** may be arranged in the same row or column, such that the first pixel circuits **30** corresponding to the sub-pixels **110** of the same color are arranged in different rows and/or different columns. Therefore, the flexibility in arranging the first pixel circuits **30** is improved.

[0041] Still referring to FIG. **4**, in the first pixel circuits **30** corresponding to the same first pixel unit **10**: one first sub-pixel circuit **310** and one third sub-pixel circuit **330** are arranged in the same column, the other first sub-pixel circuit **310** and the other third sub-pixel circuit **330** are arranged in the same column, four second sub-pixel circuits **320** are arranged in two columns, and two first sub-pixel circuits **310**, four second sub-pixel circuits **320**, and two third sub-pixel circuits **330** are arranged in at least two rows. In one embodiment, two columns of first sub-pixel circuits **310**, two columns of third sub-pixel circuits **330**, and two columns of second sub-pixel circuits **320** are sequentially arranged on one side of the corresponding first pixel unit **10**.

[0042] For example, the first pixel circuits **30** corresponding to the same first pixel unit **10** are arranged on one side of the first pixel unit **10**, and arranged in two rows and four columns. For example, from the side of the second active area AA**2** that is close to the first active area AA**1**, the first pixel circuit **30** in the first row sequentially includes the first sub-pixel circuit **310**, the third sub-pixel circuit **330**, the second sub-pixel circuit **320**, and the second sub-pixel circuit **320**, and the first pixel circuit **30** in the second row sequentially includes the third sub-pixel circuit **330**, the first sub-pixel circuit **310**, the second sub-pixel circuit **320**, and the second sub-pixel circuit **320**. This facilitates reducing the distance between the first sub-pixel circuit **310** in the first row and the first sub-pixel **111** and the distance between the third sub-pixel circuit **330** in the second row and the third sub-pixel **113**. Configuring the first sub-pixel circuit **310** in the first row to be connected to the first sub-pixel **111** in the first row in the first pixel unit **10** through the wire **40** facilitates reducing the length of the wire **40** between the first sub-pixel circuit **310** and the first sub-pixel **111**. Configuring the third sub-pixel circuit **330** in the second row to be connected to the third sub-pixel **113** in the second row in the first pixel unit **10** through the wire **40** facilitates reducing the length of the wire **40** between the third sub-pixel circuit **330** and the third sub-pixel **113**. In addition, configuring the second sub-pixel circuit **320** closest to the second sub-pixel **112** to be connected to the second sub-pixel **112** through the wire **40** further facilitates reducing the length of the wire **40** between the second sub-pixel circuit **320** and the second sub-pixel **112**.

[0043] On the basis of the above embodiments, the first pixel circuits **30** corresponding to the plurality of sub-pixels **110** in the same first pixel unit **10** are arranged in at least two adjacent rows. Such a configuration has an advantage of reducing the number of rows in which the first pixel circuits **30** corresponding to the sub-pixels **110** in the same first pixel unit **10** are arranged, so as to reduce the widths of the first pixel circuits **30** in a column direction in which the first pixel circuits are arranged, reduce the distances between the first pixel circuits **30** and the sub-pixels **110**, and then reduce the lengths of the wires **40** connected between the first pixel circuits **30** and the sub-pixels **110**.

[0044] Referring to FIG. **2** to FIG. **4**, on the basis of the above embodiments, the wire **40** may be a transparent metal wire. For example, a material of the wire **40** may include indium tin oxide (ITO) and/or indium zinc oxide (IZO). Since the first active area AA**1** has relatively high light transmittance, the wire **40** is configured as the transparent metal wire to prevent the wire **40** from affecting the light transmittance of the first active area AA**1**. The wire **40** is located in an area between adjacent second pixel units **20** or an area between a first pixel unit **10** and a second pixel unit **20** that are adjacent to each other, and the wire **40** is insulated from the first pixel circuit **30** and/or the second pixel circuit **50** to which the wire is not connected, to avoid affecting the display effect of the second pixel unit **20** in the second active area AA**2**.

[0045] For example, the display panel includes a substrate, and an array circuit layer and a first metal layer which are sequentially located on one side of the substrate. The first pixel circuit **30** and the second pixel circuit **50** are both formed in the array circuit layer. The wire **40** may be arranged in the first metal layer, to avoid affecting the display effect of the sub-pixels **110** in the first pixel unit **10** as a result of the wire **40** overlapping a signal line connected to the first pixel circuit **30** and/or the second pixel circuit **50**.

[0046] FIG. **5** is a partial enlarged view of an area B**2** in FIG. **2**. FIG. **5** shows a first row of first pixel units **10** in the first active area AA**1** and first four rows of pixel circuits (including the first pixel circuit **30** and the second pixel circuit **50**) located in the area B**2** in the second active area AA**2**. The first pixel circuits **30** in FIG. **5** may be arranged in the same way as the first pixel circuits **30** in FIG. **3**. Referring to FIG. **2**, FIG. **3**, and FIG. **5**, in one embodiment, the first active area AA**1** includes at least one row of first pixel units **10**, and first pixel circuits **30** corresponding to the sub-pixels **110** in the same row of first pixel units **10** occupy the same row.

[0047] For example, a description is made by using the first row of first pixel units **10** in the first active area AA**1** as an example. When the first pixel unit **10** includes two rows of sub-pixels **110**, the first pixel circuits **30** corresponding to the plurality of sub-pixels **110** in the same first pixel unit **10** may be arranged in two rows, for example, in a third row of pixel circuits and a fourth row of pixel circuits, and the first pixel circuits **30** corresponding to the sub-pixels **110** in the first row of first pixel units **10** are all located in the third row of pixel circuits and the fourth row of pixel circuits. Such a configuration has an advantage of connecting the first pixel circuits **30** for driving the sub-pixels **110** in the same row of first pixel units **10** to the same scan line, which, compared with the solution in which the first pixel circuits **30** for driving the sub-pixels **110** in the same row of first pixel units **10** are distributed in different rows, eliminates the need for complex wire winding when connecting the first pixel circuits **30** through the scan line, thereby simplifying the structure of the scan line and avoiding mismatched writing of a scan signal.

[0048] Referring to FIG. **2**, FIG. **3**, and FIG. **5**, in one embodiment, the first pixel circuits **30** corresponding to the sub-pixels **110** in the same row of first pixel units **10** are distributed on two sides of the row of first pixel units **10**. For example, the first pixel circuits **30** corresponding to the sub-pixels **110** in one portion of the first pixel units **10** in the same row may be distributed on one side of the row of first pixel units **10**, and the first pixel circuits **30** corresponding to the sub-pixels **110** in the other portion of the first pixel units **10** in the same row may be distributed on the other side of the row of first pixel units **10**. For example, when there are eight first pixel units **10** in each row, the first pixel circuits **30** corresponding to the first to fourth first pixel units **10** in the first row may all be arranged on a left side of the row of first pixel units **10**, with the first pixel circuits **30** corresponding to the first to fourth first pixel units **10** in the first row being sequentially arranged on the left side of the row of first pixel units **10**, and the first pixel circuits **30** corresponding to the fifth to eighth first pixel units **10** in the first row may all be arranged on a right side of the row of first pixel units **10**, with the first pixel circuits **30** corresponding to the fifth to eighth first pixel units **10** in the first row being sequentially arranged on the right side of the row of first pixel units **10**. Such a configuration has an advantage of arranging the first pixel circuits **30** corresponding to the sub-pixels **110** in the first pixel units **10** on the side adjacent to the first pixel units **10**, so as to reduce the lengths of the wires **40** connected between the first pixel circuits **30** and the sub-pixels **110**.

[0049] FIG. **6** is another partial enlarged view of the area B**2** in FIG. **2**. The first pixel circuits **30** in FIG. **6** may be arranged in the same way as the first pixel circuits **30** in FIG. **4**. Referring to FIG. **2**, FIG. **4**, and FIG. **6**, in this embodiment, the first pixel circuits **30** corresponding to the sub-pixels **110** in the same row of first pixel units **10** may be configured to occupy the same row, and the first pixel circuits **30** corresponding to the sub-pixels **110** in the same row of first pixel units **10** are distributed on two sides of the row of first pixel units **10**.

[0050] Referring to FIG. **2** to FIG. **6**, in one embodiment, in an embodiment, at least a portion of the first pixel circuits **30** may be located between adjacent second pixel circuits **50**. For example, a portion of the first pixel circuits **30** may be configured to be located between adjacent second pixel circuits **50**, or all of the first pixel circuits **30** may be configured to be located between adjacent second pixel circuits **50**, i.e., each first pixel circuit **30** is located between adjacent second pixel circuits **50**. Such a configuration has an advantage of arranging the first pixel circuits **30** in the area between adjacent second pixel circuits **50**, without the need to arrange a transition area between the first active area AA**1** and the second active area AA**2** to store the first pixel circuits **30** alone.
[0051] In one embodiment, the display panel further includes dummy pixel circuits **60** located between adjacent second pixel circuits **50**, and at least a portion of the dummy pixel circuits **60** are reused as the first pixel circuit **30**. For example, a plurality of dummy pixel circuits **60** are generally arranged between adjacent second pixel circuits **50** in the display panel. The dummy pixel circuits **60** have the same structure as the second pixel circuits **50**, except that the dummy pixel circuits **60** are generally not connected to the sub-pixels **110** and are not configured to drive the sub-pixels **110** to emit light. In the embodiment, at least a portion of the dummy pixel circuits **60** may be reused as the first pixel circuit **30**, such that driving current output terminals of the dummy pixel circuits **60** are connected to the corresponding sub-pixels **110** through the wires **40** to drive the corresponding sub-pixels **110** to emit light. In this way, there is no need to additionally arrange the first pixel circuits **30** in the display panel to drive the sub-pixels **110**, to avoid additional occupation of the space of the display panel by the first pixel circuits **30**. In addition, reusing the dummy pixel circuits **60** as the first pixel circuit **30** makes the structure of the first pixel circuit **30** consistent with that of the second pixel circuit **50**, which facilitates improving the display consistency between the first active area AA**1** and the second active area AA**2**.
[0052] For example, the second active area AA**2** includes a plurality of pixel circuit units C**1** arranged in an array. Each pixel circuit unit C**1** includes second pixel circuits **50** and dummy pixel circuits **60** located between adjacent second pixel circuits **50**. For example, each pixel circuit unit C**1** includes two rows of pixel circuits, and each row of pixel circuits sequentially includes two second pixel circuits **50**, one dummy pixel circuit **60**, and two second pixel circuits **50**. The eight second pixel circuits **50** in each pixel circuit unit C**1** are respectively configured to drive sub-pixels **110** of different colors to emit light. At least a portion of the dummy pixel circuits **60** in the pixel circuit unit C**1** may be reused as the first pixel circuit **30**. FIG. **5** and FIG. **6** both show that the dummy pixel circuits **60** in the eight pixel circuit units C**1** on two sides of the first row of first pixel units **10** are all reused as the first pixel circuit **30**. For example, the dummy pixel circuits **60** may be respectively reused as the first sub-pixel circuit **310**, the second sub-pixel circuit **320**, and the third sub-pixel circuit **330**.
[0053] FIG. **7** is a schematic diagram of a structure of another display panel according to an embodiment of the present application. FIG. **8** is a partial enlarged view of an area D**1** in FIG. **7**. Referring to FIG. **7** and FIG. **8**, in one embodiment, on the basis of the above embodiments, at least a portion of the first pixel circuits **30** may alternatively be configured to be located between adjacent second pixel circuits **50**, and at least a portion of the first pixel circuits **30** and at least a portion of the second pixel circuits **50** may be arranged in different areas in the second active area AA**2**.
[0054] Referring to FIG. **7** and FIG. **8**, for example, the first active area AA**1** includes a first active sub-area a**1** and a second active sub-area a**2**, and the second active area AA**2** includes a third active sub-area a**3** and a fourth active sub-area a**4**. The third active sub-area a**3** is adjacent to the first active sub-area a**1**, and the fourth active sub-area a**4** is adjacent to the second active sub-area a**2**. The first pixel circuit **30** corresponding to the sub-pixel **110** in the first active sub-area a**1** is located in the third active sub-area a**3**, and the first pixel circuit **30** corresponding to the sub-pixel **110** in the second active sub-area a**2** is located in the fourth active sub-area a**4**. The first pixel circuit **30** and the second pixel circuit **50** are arranged in different areas in the third active sub-area a**3**, and

the first pixel circuit **30** in the fourth active sub-area a**4** is located between adjacent second pixel circuits **50**.

[0055] For example, the first pixel circuit **30** and the second pixel circuit **50** being arranged in different areas in the third active sub-area a**3** may mean an area in the third active sub-area a**3** being used to arrange the first pixel circuit **30** only, and another area in the third active sub-area being used to arrange the second pixel circuit **50** corresponding to the second pixel unit **20** in the third active sub-area a**3**. The first pixel circuit **30** in the fourth active sub-area a**4** is located between adjacent second pixel circuits **50**. For example, when the display panel includes the dummy pixel circuits **60** located between adjacent second pixel circuits **50**, at least a portion of the dummy pixel circuits **60** in the fourth active sub-area a**4** may be reused as the first pixel circuit **30**. In the embodiment, the first pixel circuit **30** and the second pixel circuit **50** are arranged in different areas in the third active sub-area a**3**, and the first pixel circuit **30** in the fourth active sub-area a**4** is configured to be located between adjacent second pixel circuits **50**, such that the first pixel circuits **30** corresponding to the plurality of sub-pixels **110** in the first active area AA**1** are dispersed in a plurality of areas adjacent to the first active area AA**1**. This facilitates reducing the lengths of the wires connected between the first pixel circuits **30** and the corresponding sub-pixels **110**.

[0056] In one embodiment, the first active sub-area a**1** may be configured to include a fifth active sub-area a**5** and a sixth active sub-area a**6**, and the third active sub-area a**3** may be configured to include a seventh active sub-area a**7** and an eighth active sub-area a**8**. The seventh active sub-area a**7** is adjacent to the fifth active sub-area a**5**, and the eighth active sub-area a**8** is adjacent to the sixth active sub-area a**6**. The seventh active sub-area a**7** includes a first area a**71** and a second area a**72**. The first pixel circuit **30** corresponding to the sub-pixel **110** in the fifth active sub-area a**5** is located in the first area a**71**, and the second pixel circuit **50** corresponding to the sub-pixel **110** in the seventh active sub-area a**7** is located in the second area a**72**. The eighth active sub-area a**8** includes a third area a**81** and a fourth area a**82**. The first pixel circuit **30** corresponding to the sub-pixel **110** in the sixth active sub-area a**6** is located in the third area a**81**, and the second pixel circuit **50** corresponding to the sub-pixel **110** in the eighth active sub-area a**8** is located in the fourth area a**82**.

[0057] In one embodiment, the display panel further includes dummy pixel circuits **60**. At least a portion of the dummy pixel circuits **60** in the third active sub-area a**3** are reused as the first pixel circuit **30** and/or the second pixel circuit **50**. For example, the second active area AA**2** includes a plurality of pixel circuit units E arranged in an array. For example, the first area a**71** of the seventh active sub-area a**7** and the third area a**81** of the eighth active sub-area a**8** each include a first pixel circuit unit E**1**, the second area a**72** of the seventh active sub-area a**7** and the fourth area a**82** of the eighth active sub-area a**8** each include a second pixel circuit unit E**2**, and the fourth active sub-area a**4** and remaining areas of the second active area AA**2** each include a third pixel circuit unit E**3**. The first pixel circuit unit E**1** includes at least one first pixel circuit **30** and at least one dummy pixel circuit **60**. For example, the first pixel circuit unit E**1** includes two first pixel circuits **30**, one dummy pixel circuit **60**, two first pixel circuits **30**, and one dummy pixel circuit **60** that are arranged sequentially. The four first pixel circuits **30** in each first pixel circuit unit E**1** are respectively configured to drive sub-pixels **110** of different colors. For example, the four first pixel circuits **30** are sequentially configured to drive a red sub-pixel, a green sub-pixel, a blue sub-pixel, and a green sub-pixel. The dummy pixel circuit **60** in the first pixel circuit unit E**1** may be reused as the first pixel circuit **30**. The second pixel circuit unit E**2** includes at least one second pixel circuit **50** and at least one dummy pixel circuit **60**. For example, the second pixel circuit unit E**2** includes two second pixel circuits **50**, one dummy pixel circuit **60**, two second pixel circuits **50**, and one dummy pixel circuit **60** that are arranged sequentially. The four second pixel circuits **50** in each second pixel circuit unit E**2** are respectively configured to drive sub-pixels **110** of different colors. For example, the four second pixel circuits **50** are sequentially configured to drive a red sub-pixel, a green sub-pixel, a blue sub-pixel, and a green sub-pixel. The dummy pixel circuit **60** in the second pixel circuit unit E**2** may be reused as the second pixel circuit **50**. The third pixel circuit unit E**3** includes at least one second pixel circuit **50** and at least one dummy pixel circuit **60**. For example, the third pixel circuit unit E**3** includes two second pixel circuits **50**, one dummy pixel circuit **60**, and two second pixel circuits **50** that are arranged sequentially. The four second pixel circuits **50** in each third pixel circuit unit E**3** are respectively configured to drive sub-pixels **110** of different colors. For example, the four second pixel circuits **50** are sequentially configured to drive a red sub-pixel, a green sub-pixel, a blue sub-pixel, and a green sub-pixel. The dummy pixel circuit **60** in the third pixel circuit unit E**3** located in the fourth active sub-area a**4** may be reused as the first pixel circuit **30**.

[0058] Referring to FIG. **7** and FIG. **8**, in one embodiment, the first active sub-area a**1** and the second active sub-area a**2** are arranged in a first direction Y, the fourth active sub-area a**4** is located on one side of the second active sub-area a**2** that is far away from the first active sub-area a**1**, the fifth active sub-area a**5** and the sixth active sub-area a**6** are arranged in a second direction X, the seventh active sub-area a**7** and the eighth active sub-area a**8** are respectively located on either sides of the first active sub-area a**1**, the first area a**71** and the second area a**72** are arranged in the first direction Y, and the third area a**81** and the fourth area a**82** are arranged in the first direction Y, the first direction Y being perpendicular to the second direction X.

[0059] The first direction Y may be a column direction in which the sub-pixels **110** are arranged, and the second direction X may be a row direction in which the sub-pixels **110** are arranged. For example, a description is made by using, as an example, the second active sub-area a**2** being located on a lower side of the first active sub-area a**1** and the fifth active sub-area a**5** being located on a left side of the sixth active sub-area a**6**. In this case, the seventh active sub-area a**7** is located on a left side of the fifth active sub-area a**5**, the eighth active sub-area a**8** is located on a right side of the sixth active sub-area a**6**, and the fourth active sub-area a**4** is located on a lower side of the second active sub-area a**2**. Such a configuration has an advantage of arranging the first pixel circuits **30** corresponding to the sub-pixels **110** in the first active area AA**1** in three directions on an outer side of the first active area AA**1**, thereby improving the flexibility in arranging the first pixel circuits **30**. Moreover, arranging the first pixel circuits **30** corresponding to the sub-pixels **110** in the first active area AA**1** on a left side, a right side, and a lower side of the first active area AA**1** eliminates the need to arrange the first pixel circuits **30** on an upper side of the first active area AA**1**, i.e., on the side of the first active area AA**1** that is close to the non-active area NAA. Therefore, in the first direction Y, the area of the second active area AA**2** between the upper side of the first active area AA**1** and the non-active area NAA can be smaller, or the first active area AA**1** may even be arranged adjacent to the non-active area NAA, which can reduce the distance between the first active area AA**1** and the non-active area NAA, and then reduce the distance between an under display camera area and the non-active area NAA, thereby improving the user experience. In this case, there is no transition area at the periphery of the first active area AA**1**.

[0060] As an optional embodiment of the present application, the first active area AA**1** may be configured to include eight rows and eight columns of first pixel units **10**, the fifth active sub-area a**5** may be configured to include first to fourth columns of the first to fourth rows of first pixel units **10**, the sixth active sub-area a**6** may be configured to include fifth to eighth columns of the first to fourth rows of first pixel units **10**, and the second active sub-area a**2** may be configured to include fifth to eighth rows of first pixel units **10**. Correspondingly, the first pixel circuit **30** corresponding to the first pixel unit **10** in the fifth active sub-area a**5** may be arranged in the first area a**71** of the seventh active sub-area a**7**, and the second pixel circuits **50** corresponding to the four rows and four columns of second pixel units **20** in the seventh active sub-area a**7** may be arranged in the second area a**72** of the seventh active sub-area a**7**. The first pixel circuit **30** corresponding to the first pixel unit **10** in the sixth active sub-area a**6** may be arranged in the third area a**81** of the eighth active sub-area a**8**, and the second pixel circuits **50** corresponding to the four rows and four columns of second pixel units **20** in the eighth active sub-area a**8** may be arranged in the fourth area a**82** of the eighth active sub-area a**8**. The fourth active sub-area a**4** may include four rows and eight columns of second pixel units **20**. The second pixel circuits **50** corresponding to the second pixel units **20** in the fourth active sub-area a**4** and the first pixel circuits **30** corresponding to the first pixel units **10** in the second active sub-area a**2** may both be arranged in the fourth active sub-area a**4**.

[0061] In one embodiment, in other embodiments, it is also possible to arrange at least a portion of the first pixel circuits **30** and at least a portion of the second pixel circuits **50** in different areas in the second active area AA**2**, instead of arranging the first pixel circuits **30** between adjacent second pixel circuits **50**. In actual application, it is possible to choose, according to requirements, whether to arrange at least a portion of the first pixel circuits **30** and at least a portion of the second pixel circuits **50** in different areas in the second active area AA**2** and/or whether to arrange at least a portion of the first pixel circuits **30** between adjacent second pixel circuits **50**.

[0062] Referring to FIG. **2** or FIG. **8**, on the basis of the above embodiments, the first active area AA**1** and the second active area AA**2** may have the same or different sub-pixel densities. The sub-pixel density is the number of sub-pixels **110** per unit area of the active area (pixels per inch, PPI). The sub-pixel density of the first active area AA**1** and the sub-pixel density of the second active area AA**2** may be set according to requirements, which are not limited in this embodiment.

[0063] On the basis of the above embodiments, the sub-pixels in the first active area AA**1** and the sub-pixels in the second active area AA**2** are arranged in the same way. For example, the sub-pixels **110** of different colors in the first pixel unit **10** in the first active area AA**1** may be arranged in the same way as the sub-pixels **110** of different colors in the second pixel unit **20** in the second active area AA**2**. For example, the arrangement of the sub-pixels in the first active area AA**1** and the second active area AA**2** may be the arrangement of the sub-pixels **110** of different colors shown in FIG. **3** and FIG. **4**.

[0064] On the basis of the above embodiments, the first pixel circuit **30** and the second pixel circuit **50** have the same structure. Such a configuration has an advantage of improving the display consistency between the first active area AA**1** and the second active area AA**2**, and simplifying the manufacturing process of the display panel without the need to design pixel circuits with different structures for the first active area AA**1** and the second active area AA**2**. In addition, the first active area AA**1** and the second active area AA**2** may further be configured to support different gamma algorithms to respectively adjust the display effects in the first active area AA**1** and the second active area AA**2**, thereby improving the uniformity of the overall display brightness of the display panel.

[0065] FIG. **9** is a schematic diagram of a structure of another display panel according to an embodiment of the present application. Referring to FIG. **2** to FIG. **9**, on the basis of the above embodiments, the display panel further includes scan lines **70**. The first pixel circuits **30** for driving the sub-pixels **110** in the same row of first pixel units **10** are connected to the same scan line **70**. For example, the first pixel circuit **30** includes a switching transistor. Each scan line **70** is connected to a gate electrode of the switching transistor in the corresponding first pixel circuit **30** to supply a scan signal to the gate electrode of the switching transistor, so as to control turning-on and turning-off of the switching transistor. The first pixel circuits **30** for driving the sub-pixels **110** in all rows of first pixel units **10** are respectively connected to the corresponding scan lines **70**, and the first pixel circuits **30** for driving the sub-pixels **110** in the same row of first pixel units **10** are connected to the same scan line **70**, such that row-by-row scanning of the first pixel units **10** in the first active area AA is implemented.

[0066] In one embodiment, the scan line **70** connected to the first pixel circuits **30** is located outside the first active area AA**1**; or the scan line **70** connected to the first pixel circuits **30** passes through the first active area AA**1**, and the scan line **70** located in the first active area AA**1** is a transparent metal wire.

[0067] For example, when the first pixel circuits **30** corresponding to the sub-pixels **110** in the same row of first pixel units **10** are distributed on two sides of the row of first pixel units **10**, the scan line **70** needs to be connected to each of the first pixel circuits **30** located on the two sides of the row of first pixel units **10**. In an embodiment, the scan lines **70** may be configured to be located outside the first active area AA**1**. For example, the scan lines **70** may extend from a left edge of the second active area AA**2** toward the first active area AA**1** in the row direction in which the first pixel circuits **30** are arranged, and perform, at a position close to the first active area AA**1** and at an outer edge of the first active area AA**1**, wire winding from one side of the first active area AA**1** to the other side thereof, and then continue to extend from an edge of the first active area AA**1** toward a right edge of the second active area AA**2** in the row direction in which the first pixel circuits **30** are arranged, such that the scan lines **70** are arranged to be wound at an outer periphery of the first active area AA**1**. This avoids affecting the display effect in the first active area AA**1** as a result of the scan lines **70** passing through the first active area AA**1**. In another embodiment, the scan lines **70** connected to the first pixel circuits **30** may be configured to pass through the first active area AA**1**. For example, the scan lines **70** may be configured to extend from a left edge of the second active area AA**2** through the first active area AA**1** toward a right edge of the second active area AA**2** in the row direction in which the first pixel circuits **30** are arranged, such that the scan lines **70** are connected to the first pixel circuits **30** located on two sides of the first active area AA**1**. The scan lines **70** each include a portion of the scan line located in the first active area AA**1**. The portion of the scan line **70** is a transparent metal wire. Such a configuration has an advantage that the scan lines **70** can extend in the same direction, which facilitates simplifying the structure of the scan lines **70**. Moreover, since the scan lines **70** in the first active area AA**1** are transparent metal wires, the scan lines **70** in the first active area AA**1** do not affect the display effect of the first active area AA**1**.

[0068] It should be noted that FIG. **9** shows that a portion of the scan lines **70** connected to the first pixel circuits are located outside the first active area AA**1**, and a portion of the scan lines **70** pass through the first active area AA**1**, the scan lines **70** located in the first active area AA**1** being transparent metal wires. In other embodiments, all of the scan lines **70** connected to the first pixel circuits may be configured to be located outside the first active area AA**1**, or all of the scan lines **70** connected to the first pixel circuits may be configured to pass through the first active area AA**1**, the scan lines **70** located in the first active area AA**1** being transparent metal wires.

[0069] It should be noted that the first active area AA**1** being in the shape of a rectangle and the first active area AA**1** being located near the top of the active area AA as a whole is used as an example for illustration in FIG. **2**, FIG. **7**, and FIG. **9**. In other embodiments, the first active area AA**1** is not limited to be in the shape of a rectangle, and the first active area AA**1** may be in other shapes, such as a circle, a droplet, or a “U” shape, and the first active area AA**1** may alternatively be arranged at other positions of the active area AA. The shape and position of the first active area AA**1** are not limited in this embodiment of the present application.

[0070] An embodiment of the present application further provides a display apparatus. The display apparatus includes a display panel according to any one of the above embodiments. The display apparatus may be an apparatus with a display function, such as a mobile phone, a desktop computer, a notebook computer, or a tablet computer. The display apparatus according to the embodiments of the present application includes a display panel according to any one of the above embodiments, and thus has the corresponding functional structures of the display panel. Details are not described herein.

Claims

1. A display panel, which has a first active area and a second active area, wherein a light transmittance of the first active area is greater than a light transmittance of the second active area, the display panel comprising: a plurality of first pixel units located in the first active area, the first pixel unit comprising a plurality of sub-pixels, wherein the sub-pixel comprises a first electrode and a second electrode opposite the first electrode; and a plurality of first pixel circuits located in the second active area, the first pixel circuit being connected to the first electrode of the corresponding sub-pixel in the first pixel unit through a wire, wherein the plurality of first pixel circuits are arranged in an array, and the first pixel circuits corresponding to the plurality of sub-pixels in a same first pixel unit are arranged in at least two rows and in a number of columns less than a total number of sub-pixels in the first pixel unit.

2. The display panel according to claim 1, wherein the first pixel unit comprises m sub-pixels of the same color, and the m sub-pixels of the same color in the same first pixel unit are driven by n first pixel circuits, wherein n and m are both positive integers greater than or equal to 1, and $n \leq m$.

3. The display panel according to claim 2, wherein for $n=m=1$, each first pixel circuit is connected to one corresponding sub-pixel through the wire; for $n=1$ and $m>1$, the first electrodes of the m sub-pixels of the same color in the same first pixel unit are electrically connected to each other, and are electrically connected to the corresponding first pixel circuit through a same wire; and for $n>1$ and $m>1$, n first pixel circuits corresponding to the m sub-pixels of the same color in the same first pixel unit are electrically connected to each other, and the first electrodes of the m sub-pixels of the same color are electrically connected to each other, and are electrically connected to any first pixel circuit among the n corresponding first pixel circuits through a same wire.

4. The display panel according to claim 3, wherein for $n>1$ and $m>1$, the m sub-pixels of the same color in the same first pixel unit are connected to the closest first pixel circuit among the n corresponding first pixel circuits through the wire.

5. The display panel according to claim 1, wherein at least part of the plurality of sub-pixels in the first pixel unit have different colors, and there is at least one sub-pixel of each color; and the first pixel circuits corresponding to the sub-pixels of the same color in the same first pixel unit are arranged in a same row; or the first pixel circuits corresponding to the sub-pixels of the same color in the same first pixel unit are arranged in a same column; or the first pixel circuits corresponding to the sub-pixels of the same color in the same first pixel unit are arranged in at least one of different rows and different columns.

6. The display panel according to claim 1, wherein the plurality of sub-pixels of the first pixel unit comprises a plurality of first sub-pixels, a plurality of second sub-pixels, and a plurality of third sub-pixels, the first sub-pixels, the second sub-pixels, and the third sub-pixels having different colors, and the same first pixel unit comprises two first sub-pixels and two third sub-pixels, as well as four second sub-pixels; the first sub-pixels, the second sub-pixels, and the third sub-pixels are arranged in an array in the first pixel unit, a first sub-pixel of the first sub-pixels, a second sub-pixel of the second sub-pixels, a third sub-pixel of the third sub-pixels, and a second sub-pixel of the second sub-pixels being sequentially arranged in a first row of the first pixel unit, and a third sub-pixel of the third sub-pixels, a second sub-pixel of the second sub-pixels, a first sub-pixel of the first sub-pixels, and a second sub-pixel of the second sub-pixels being sequentially arranged in a second row of the first pixel unit; and the first sub-pixel is a red sub-pixel, the second sub-pixel is a green sub-pixel, and the third sub-pixel is a blue sub-pixel.

7. The display panel according to claim 6, wherein the first pixel circuit comprises a plurality of first sub-pixel circuits, a plurality of second sub-pixel circuits, and a plurality of third sub-pixel circuits, the two first sub-pixels in the same first pixel unit being driven by two first sub-pixel circuits, the two third sub-pixels in the same first pixel unit being driven by two third sub-pixel circuits, and the four second sub-pixels in the same first pixel unit being driven by four second sub-pixel circuits; and in the first pixel circuits corresponding to the same first pixel unit: two first sub-pixel circuits are arranged in a column, two third sub-pixel circuits are arranged in a column, four second sub-pixel circuits are arranged in two columns, and the two first sub-pixel circuits, the four second sub-pixel circuits, and the two third sub-pixel circuits are arranged in at least two rows; and the column of the two first sub-pixel circuits, the two columns of the four second sub-pixel circuits, and the column of the two third sub-pixel circuits are sequentially arranged on one side of the

corresponding first pixel unit.
8. The display panel according to claim 6, wherein the first pixel circuit comprises a plurality of first sub-pixel circuits, a plurality of second sub-pixel circuits, and a plurality of third sub-pixel circuits, the two first sub-pixels in the same first pixel unit being driven by two first sub-pixel circuits, the two third sub-pixels in the same first pixel unit being driven by two third sub-pixel circuits, and the four second sub-pixels in the same first pixel unit being driven by four second sub-pixel circuits; and in the first pixel circuits corresponding to the same first pixel unit: one of the two first sub-pixel circuits and one of the two third sub-pixel circuits are arranged in a column, another one of the two first sub-pixel circuits and another one of the two third sub-pixel circuits are arranged in a column, four second sub-pixel circuits are arranged in two columns, and the two first sub-pixel circuits, the four second sub-pixel circuits, and the two third sub-pixel circuits are arranged in at least two rows; and the column of the one of the two first sub-pixel circuits and the one of the two third sub-pixel circuits, the column of the another one of the two first sub-pixel circuits and the another one of the two third sub-pixel circuits, and the two columns of the four second sub-pixel circuits are sequentially arranged on one side of the corresponding first pixel unit.
9. The display panel according to claim 1, wherein the first pixel circuits corresponding to the plurality of sub-pixels in a same first pixel unit are arranged in at least two adjacent rows.
10. The display panel according to claim 1, wherein the first active area comprises at least one row of first pixel units, and the first pixel circuits corresponding to the sub-pixels in the same row of first pixel units occupy the same row.
11. The display panel according to claim 10, wherein the first pixel circuits corresponding to the sub-pixels in the same row of first pixel units are distributed on two sides of the row of first pixel units.
12. The display panel according to claim 1, wherein the wire is a transparent metal wire; and a material of the wire comprises at least one of indium tin oxide and indium zinc oxide.
13. The display panel according to claim 1, further comprising a plurality of scan lines, wherein the first pixel circuits for driving the sub-pixels in the same row of first pixel units are connected to a same scan line.
14. The display panel according to claim 13, wherein the scan line connected to the first pixel circuits is located outside the first active area; or the scan line connected to the first pixel circuits passes through the first active area, and at least part of the scan line located in the first active area is a transparent metal wire.
15. The display panel according to claim 1, further comprising a plurality of second pixel units and a plurality of second pixel circuits, wherein the second pixel units and the second pixel circuits are both located in the second active area, the second pixel unit comprising a plurality of sub-pixels, and the second pixel circuits being configured to drive the corresponding sub-pixels in the second pixel unit.
16. The display panel according to claim 15, wherein the display panel meets at least one of the following: at least a portion of the first pixel circuits being located between adjacent second pixel circuits; and at least a portion of the first pixel circuits and at least a portion of the second pixel circuits being arranged in different areas in the second active area.
17. The display panel according to claim 16, wherein each first pixel circuit is located between adjacent second pixel circuits; and the display panel further comprises a plurality of dummy pixel circuits located between adjacent second pixel circuits, and at least part of the dummy pixel circuits are reused as the first pixel circuits.
18. The display panel according to claim 16, wherein the first active area comprises a first active sub-area and a second active sub-area, and the second active area comprises a third active sub-area and a fourth active sub-area, the third active sub-area being adjacent to the first active sub-area, and the fourth active sub-area being adjacent to the second active sub-area; the first pixel circuit corresponding to the sub-pixel in the first active sub-area is located in the third active sub-area, and

the first pixel circuit corresponding to the sub-pixel in the second active sub-area is located in the fourth active sub-area; and the first pixel circuit and the second pixel circuit are arranged in different areas in the third active sub-area, and the first pixel circuit in the fourth active sub-area is located between adjacent second pixel circuits.

19. The display panel according to claim 18, wherein the first active sub-area comprises a fifth active sub-area and a sixth active sub-area, and the third active sub-area comprises a seventh active sub-area and an eighth active sub-area, the seventh active sub-area being adjacent to the fifth active sub-area, and the eighth active sub-area being adjacent to the sixth active sub-area; the seventh active sub-area comprises a first area and a second area, the first pixel circuit corresponding to the sub-pixel in the fifth active sub-area being located in the first area, and the second pixel circuit corresponding to the sub-pixel in the seventh active sub-area being located in the second area; the eighth active sub-area comprises a third area and a fourth area, the first pixel circuit corresponding to the sub-pixel in the sixth active sub-area being located in the third area, and the second pixel circuit corresponding to the sub-pixel in the eighth active sub-area being located in the fourth area; the display panel further comprises dummy pixel circuits, wherein at least a portion of the dummy pixel circuits in the third active sub-area are reused as at least one of the first pixel circuit and the second pixel circuit; at least a portion of the dummy pixel circuits in the fourth active sub-area are reused as the first pixel circuit; and the first active sub-area and the second active sub-area are arranged in a first direction, the fourth active sub-area is located on one side of the second active sub-area that is far away from the first active sub-area, the fifth active sub-area and the sixth active sub-area are arranged in a second direction, the seventh active sub-area and the eighth active sub-area are respectively located on either sides of the first active sub-area, the first area and the second area are arranged in the first direction, and the third area and the fourth area are arranged in the first direction, the first direction being perpendicular to the second direction.

20. A display apparatus, comprising a display panel according to claim 1.